



Effects of soil treatments with entomopathogenic fungi on soil dwelling non-target arthropods at a commercial olive orchard

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ABSTRACT

Recent studies have shown the potential of entomopathogenic fungi (EF) for the biocontrol of tephritid puparia in soil treatments beneath the tree canopy. The soil is the natural ecosystem of these fungi, and the soil environment provides protection against UV. Furthermore, the fungi may recycle in the soil, providing a long term control of the puparia. However, such soil applications could have a negative impact on non-target soil arthropod fauna. In particular, ants play an important role in olive grove soil fauna because they are biological indicators of the soil condition. Thus, the aims of this study were to (i) determine the persistence of *Metarhizium anisopliae* EAMa 01/58-Su isolate in the soil when applied beneath olive trees for controlling olive fly puparia, (ii) elucidate its possible effect on non-target soil dwelling arthropod communities and (iii) evaluate the effect of soil treatment with *M. anisopliae* and *Beauveria bassiana* on *Tapinoma nigerrimum* colonies. Our results indicate that the soil ecosystem favors the persistence of this autochthonous isolate, maintaining levels of 10^5 CFUg⁻¹ of soil. This level allows long-term protection of the crop against the olive fly puparia. Furthermore, in pitfall traps, no infected insects were found in field as a result of the fungal treatment. The most abundant arthropods trapped were formicidae species. In laboratory assays, no significant differences in mortality were found between fungal treatments and control. The mean mortality values were between 41.0% and 64.7%. However, a significant reduction in average survival time was observed with 16.9 days in control to 15.6 days in the treatment with *B. bassiana* and 14.8 days in the treatment with *M. anisopliae*. Furthermore, there were no significant differences in ant activity before and after fungal treatment. The soil samples were also evaluated to determinate if these ants distribute conidia from treated soils to non-treated soils without being able to notice the presence of EF.

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1. Introduction

The olive fly *Bactrocera oleae* Gmelin is the main pest of the olive crop and it is largely responsible for around 60% of the losses due to insects (Mazomenos et al., 1997). The predominant method of olive fly control has been through the use of conventional insecticides targeting adults, but the continued use of such products has been questioned in the last years. It is well known that the wasteful over-use of insecticides (i.e. dimethoate) poses a threat for the public health and it increases environmental side effects such as insecticide resistance of target pests (Skouras et al., 2007), which is detrimental to the natural enemies, with the subsequent appearance of secondary pests (Katsoyannos, 1985; Santos et al., 2010). For this reason, there are considerable efforts to develop alternative for olive fruit fly control, promoting agricultural strategies of

low environmental impact, such as integrated production and organic farming. As an alternative to chemical control there is resurgence in the use of microbial insecticides for biological control of tephritids, particularly entomopathogenic fungi (EF). They infect their host through the cuticle, so they must not be ingested to be effective, offering an excellent opportunity for a successful control of soil-dwelling stages of insects. The soil is the natural ecosystem of EF as its environment provides protection against UV along with optimal conditions of temperature and moisture. Furthermore, the fungi may survive in the soil through recycling in insects or roots (Leger, 2008), may provide a long-term strategy for the puparia control. Recent research shows the potential of EF in soil inoculation targeting pupariating larvae and puparia (Ekesi et al., 2003; Quesada-Moraga et al., 2006).

Although at isolate level, entomopathogenic mitosporic ascomycetes may have different levels of specificity, at species level, these fungi show low specificity for their insect host (Hesketh et al., 2010; Jaronski, 2010), so these soil applications could have

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a negative impact on beneficial insects, such as parasitoids and predators, which regulate phytophagous population (Ekesi et al., 2005; Pearson and Callaway, 2005). The best way to address this is to perform impact assessment on the olive orchard arthropod fauna. Ants are the most abundant element in olive orchards and play an important role in olive grove soil fauna because they are biological indicators of the soil condition (Redolfi et al., 2005). Several studies have shown the susceptibility of various ant species to EF (Santos et al., 2007a), but no works have studied ants in the habitat conditions that we are presenting here. *Tapinoma nigerrimum* Nylander is considered one of the most abundant ant species in olive groves (Redolfi et al., 1999). It is a generalist species used in activity pattern studies (Lopez et al., 1997; Redolfi et al., 2002), and it nests in the soil where EF is present. This species is considered omnivorous, collecting sugar secretions exuded by insects but also preying on some insects, some of which are pests (Morris et al., 2002).

The aim of this study was to determine the side effect of soil treatment with *Metarhizium anisopliae* (Metsch.) Sorok. and *Beauveria bassiana* (Balsamo) Vuill. on soil dwelling non-target arthropods at a commercial olive orchard, with an emphasis on ants. This paper describes pre- and post-contact responses of ants to EF in terms of activity as well as their mortality and fungi transmission potential.

2. Materials and methods

2.1. Fungal strains and cultivation

Metarhizium anisopliae strain EAMa 01/58-Su from the culture collection at the Department of Agricultural and Forestry Sciences and Resources (AFSR) of the University of Cordoba (Spain) was originally isolated from the soil of a wheat crop at Hinojosa del Duque (Cordoba, Spain). *Beauveria bassiana* strain EABb 01/103-Su was isolated from a forest soil in Sevilla (Spain). *Metarhizium anisopliae* EAMa 01/58-Su strain was selected for evaluating the effect of soil treatment on soil dwelling non-target arthropods while the combination of the *M. anisopliae* strain and the *B. bassiana* EABb 01/103-Su strain were used for studying the effects of soil treatments on *Tapinoma nigerrimum* colonies under laboratory conditions. These strains were selected because they have been shown in our previous work to be the most virulent against both adult and puparia of *C. capitata* and *B. oleae* (Quesada-Moraga et al., 2006; Garrido-Jurado, 2008). Slant monoconidial cultures of the strains were grown on Malt Agar (MA) at 25 °C in darkness, then were lyophilized in the LyoQuest laboratory freeze dryer (Telstar, Barcelona, Spain), and stored at –80 °C (Humber, 1996).

Both strains were cultured on rice in polypropylene bags. After fungal growth, conidia were collected by sieving and suspended in sterile aqueous 0.1% Tween 80 solution. The conidial concentration was quantified in a Malassez chamber and adjusted to 1.0×10^8 conidia ml⁻¹. The viability of the conidia was verified before the preparation of suspensions by germinating tests in liquid Czapek-Dox broth plus 1% (w/v) yeast extract medium. In all the experiments, germination rates were higher than 90%.

2.2. Effects of field soil treatment on soil dwelling non-target arthropods

2.2.1. Experimental site

The study was conducted at Antequera, Málaga (37° 01'N, 4° 41'W) on an organic olive orchard with spontaneous plant cover in July and August 2007 and 2009 with the same plots in both years. The mean air temperatures were 24.0 °C in 2007 and 27.5 °C in 2009, while the soil temperature was constant for both

years (the temperature ranged from 26–29 °C). The olive orchard was 400 ha with trees spaced 10 m × 10 m and contained the olive varieties Picual, Hojiblanca, Arbequina and Cornicabra. The study site was divided into two square 1 ha sub-fields approximately 400 m apart. One sub-field was the fungi treated plot and the other was the control plot (water treated). Each plot has 100 drip-irrigated olive cv. 'Hojiblanca'.

2.2.2. Monitoring of fungus persistence

The soil beneath the canopy of the trees was sprayed with a suspension of conidia of *M. anisopliae* EAMa 01/58-Su strain at the rate of 2.5×10^{11} conidia per ha during the second week of June in both years. Before fungal treatment, 10 completely randomized soil samples were collected to determine the natural presence of entomopathogenic fungi in the soil. Finally, the *Galleria* Bait Method was applied to each soil sample to isolate entomopathogenic fungi (Zimmermann, 1986).

After spraying with the treatment, 10 trees were randomly selected to evaluate the inoculum presence in soil. For this purpose soil samples were collected beneath the canopy following the same procedure as mentioned above at 0, 7, 14, 21, 28, 35, 42, 49 and 56 days after treatment was applied. To assess the conidial density in each sample, the number of colony forming units (CFU) per gram of dry soil was determined by Sabouraud glucose agar chloramphenicol (0.5 g/l) in Petri dishes (Goettel and Inglis, 1996). One gram of the homogenized sample of soil was added to 9 ml of sterile distilled water, and shaken for 20–60 min. After homogenization, aliquots of 100 µl were spread onto the medium. In some cases, it was necessary to dilute the soil solution before spreading.

2.2.3. Collection of arthropods

Arthropod population in each sub-field was sampled using pitfall traps (75 mm diameter by 100 mm deep) every two weeks after the initial treatment, during July and August 2007 and 2009 with a total of four samplings per year. Twenty pitfall traps were randomly located on each sub-field. The traps were buried with the lip of the container flush with the soil surface. Each trap contained 125 ml soaped water to keep the insects inside. Traps were removed after 2 days and carried to the laboratory for counting and identification of insects. The insects were identified to the taxonomic level of order and the total number of each taxon recorded (Barrientos, 2004). However, the family Formicidae was identified to the taxonomic level of genus due to its abundance. Determination of possible fungal infection in the collected specimens was performed with a humid chamber. Before being placed in the humid chamber the arthropod specimens were washed off with sterilized water to remove soil particles. They were then placed in 1% sodium hypochlorite for 3 min and rinsed twice with sterilized water. After that the insects were placed on a damp sterilized filter paper in Petri dishes that were kept at 25 °C. Fungi were prepared with lactophenol cotton blue and they were examined by contrast microscopy (magnification, 100×).

2.3. Effects of soil treatments on *Tapinoma nigerrimum* colonies under laboratory conditions

2.3.1. Sampling of ants

Twenty-eight nests of *Tapinoma nigerrimum* were sampled in "Arenales de San Pedro" Granada (37° 18'N, 3° 39'W). More than 100 workers were captured per nest and nests were considered different when separated by more than 20 m.

Ants were transferred to plastic cages ringed with Fluon® to prevent their escape. They were maintained in these cages for one month at insectary conditions (25 ± 2 °C, 60 ± 5% RH and 16 h/day length) to become acclimatized to the new environment. Honey was used as the standard diet twice a week, and lacewings

larvae, aphids or *Ephestia kuehniella* (flour moth) eggs were provided once a week as a protein source.

The ants were placed in experimental nests to perform the bioassay. These ones consisted in two clear plastic cages linked by a plastic tube. The smaller cage was considered the artificial nest (Fig. 1) and was supplied with water. The bigger cage was considered the feeding chamber, like their external world, and it contained 150 g of soil (modified from Redolfi et al., 2002).

2.3.2. Experimental procedures

2.3.2.1. Pathogenicity of *B. bassiana* and *M. anisopliae* against *T. nigerrimum*. Ants from 20 different nests were exposed to the fungal entomopathogens *B. bassiana* and *M. anisopliae* ($n = 10$) in the experimental nests. Eight different nests served as the control. In each treatment group, the nests were treated with 13.5 ml of a 1.0×10^8 conidia/ml fungal suspension adjusted by diluting conidia with 0.1% aqueous solution of Tween 80. In the control group, the nests were treated with the same volume of a sterile aqueous solution of 0.1% Tween 80. The experiment was continued for 20 days. Dead ants were removed daily and they were placed in humid chamber. They were then placed on sterile wet filter paper in sterile petri dishes that were sealed with Parafilm and kept at 25 °C. They were later inspected for development of mycosis on the cadavers.

2.3.2.2. Activity of *T. nigerrimum*. Ant activity level is often an important factor when examining the effects of changes in ecosystems. These changes in spatial pattern resulting from altered behavior of ants are used as biological indicators. The activity of the ants was observed before and after the treatment. The actual number of ants going across the link tube of both sides of the experimental nest was observed for 5 min every 12 h. Activity measures taken place in the middle of the tube represented by a circle in Fig. 1. This method was continued for 11 days before the treatment and 11 days after the treatment.

2.3.2.3. Fungal dispersal by *T. nigerrimum*. The potential of these ants to distribute conidia from treated soil to non-treated soil was evaluated. An annex with sterile soil was linked to the treated experimental nests, and ants were allowed to cross from one side to the other. Soil samples from the non-treated nest were collected every 2 days and they were examined by the CFU method to determine the presence of fungi, as it was explained in 2.2.2.

2.4. Statistical analysis

Values of CFU g⁻¹ soil were Log₁₀ transformed make the data linear. After confirming that all data were normally distributed, a one-way analysis of covariance (ANCOVA) was performed to com-

pare the linear regressions of the inoculum density decline in soil (Rumbos et al., 2008). The log transformation of CFU in 2009 was used as the dependent variable and the log transformation of CFU in 2007 was used as the covariate.

The Kruskal–Wallis one-way non-parametric AOV analysis for pitfall traps data in both treated and control fields was performed using the program Statistix 9.0 (Analytical software 2008). Likewise, the mortality data were analyzed using one-way analysis of variance (ANOVA) while Tukey's honest significant difference (HSD) test was used to compare means.

The cumulative mortality response across the assessment period was analyzed using the Kaplan–Meier survival analysis. A general linear model for repeated measurement was used to analyze the activity of *T. nigerrimum* before and after treatment. These analyses were performed using the SPSS 15.0 for Windows (SPSS Inc. 2006).

3. Results

3.1. Monitoring of fungus persistence

Two autochthonous fungal species were obtained prior to treatment through the *Galleria* Bait Method. While *B. bassiana* was obtained from all the analyzed soil samples, *M. anisopliae* was found only in one soil sample at 10⁴ CFU g⁻¹ of soil, coexisting with *B. bassiana*.

During both years of the study, EAMa 01/58-Su densities decreased over time after the treatment application (Fig. 2). The reduction in CFU numbers compared to initial densities ranged from 37.54% in 2007 to 28.11% in 2009, 56 days after the treatment application. In 2007, the regression coefficient describing the linear decline of EAMa 01/58-Su density over time was -0.0025 (Fig. 2). However, in 2009 the linear decline of the EAMa 01/58-Su density in the soil over time was described by the regression coefficient of -0.0022 . The analysis of covariance revealed no significant differences ($F_{1,8} = 0.95$, $P = 0.362$) between the regression coefficients of the linear regressions describing the dynamics of EAMa 01/58-Su isolate in the different years of treatment.

3.2. Effects of field soil treatment on soil dwelling non-target arthropods

The total numbers of arthropods collected during each year of the study were 6766 and 5572 for 2007 and 2009, respectively. They were divided into 16 orders, 12 of which belonged to the Insecta class. Hymenoptera (particularly Formicidae) was the dominant taxa in terms of number of individuals from all subfields in both years. A total of 63.61% of all organisms captured in the first year and 88.48% in the second year belonged to this taxa. *Cataglyphis*

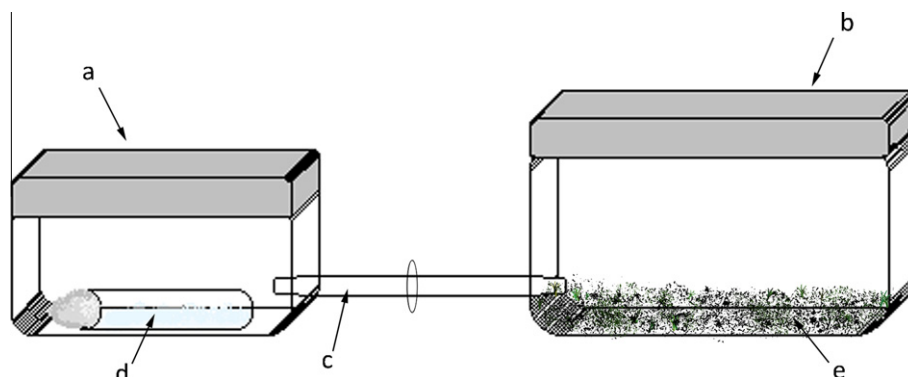


Fig. 1. Experimental nest (a) nest (80 × 150 × 80 mm), (b) feeding chamber (100 × 200 × 120 mm), (c) linker tube (80 × 10 mm), (d) water, and (e) 150 g soil.

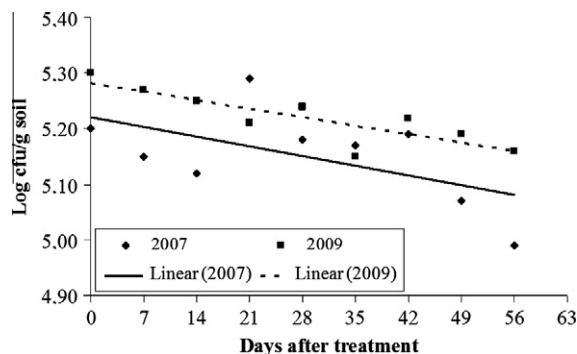


Fig. 2. Conidial persistence of *Metarhizium anisopliae* strain EAMa 01/58-Su after soil application in the olive orchard in years 2007 and 2009 over 56 days. The inoculum recovered is expressed as colony forming units (CFU).

and *Pheidole* account for 31.61–22.56% and 17.47–43.02%, respectively, of the Formicidae captures, followed by *Messor* with 13.82–19.26% (Table 1).

The Acariformes, Cephalostigmata, Orthoptera, Embioptera, Blattodea, Mantodea, Thysanoptera, Neuroptera, and Lepidoptera orders were excluded from the statistical analysis due to the lower catches. These orders were found in very limited numbers and were therefore not considered representative.

For the number of captures, the Kruskal–Wallis one-way analysis showed a significant difference between the years of treatment ($H = 67.14$, $P < 0.001$). A decrease in total number of individuals (from 6766 to 5572) was observed during the second year throughout the entire experimental field.

Furthermore, there were significant differences between treatments (2007: $H = 8.03$, $P = 0.004$ and 2009: $H = 7.17$, $P = 0.007$). In both years, the total number of individuals caught in the fungal treatment sub-field was higher than the total number of individuals caught in the control sub-field. There were significant differences between the orders trapped each year ($H = 93.52$, $P < 0.001$ in 2007 and $H = 84.11$, $P < 0.001$ in 2009). In 2007, one dominant

order in addition to ants was Isopoda. A total of 27.5 and 52.9 individuals were captured per trap for the control and treated sub-field, respectively. The remaining orders used for the statistical analysis appeared infrequently. Only 1.9–4.0 individuals per trap were captured in the control and 2.3–6.7 in the treated sub-field. Two dominant taxa (>3.0 individuals per trap in both treatments) appeared throughout both experimental sub-fields during the second year. These were Araneae and Hemiptera.

The Formicidae family was analyzed separately because of the higher catch. Only *Messor*, *Plagiolepis*, *Cataglyphis* and *Pheidole* genera were considered for this analysis as the catches of the remainder were erratic. There were significant differences in the Kruskal–Wallis one-way analysis between the years ($H = 5.23$, $P < 0.001$). In 2007, a total of 4293 ants were collected throughout all the experimental subfields. However, an increase in the total number of ants (4841) was observed during the second year.

Furthermore, there were significant differences between the treatments in 2007 ($H = 7.03$, $P = 0.008$). The total number of ants collected in the fungal treatment sub-field was higher than the total number of ants collected in the control sub-field. There were significant differences between the genera trapped each year (2007: $H = 99.65$, $P < 0.001$ and 2009: $H = 120.03$, $P < 0.001$). In 2007, one genus was dominant, which was *Cataglyphis* with 41.1 individuals per trap in the control and 65.9 in the treated sub-field. During 2007, the remaining genus appeared at minor frequencies. In 2009, *Pheidole* was the dominant genus with 70.6 ants per trap in the control and 49.3 in the treated sub-field. As a result of the high presence of this genus there were no significant differences between the control and the treated sub-fields.

3.3. Effects of soil treatments on *Tapinoma nigerrimum* colonies under laboratory conditions

3.3.1. Pathogenicity of *B. bassiana* and *M. anisopliae* against *T. nigerrimum*

No significant differences were found on workers mortality between fungal treatments and control (Table 2, $F_{2,27} = 2.34$, $P = 0.13$).

Table 1
Pitfall captures of arthropods taxa from an organic olive orchard at Antequera (Málaga).

Taxa collected			2007			2009		
Order	Family	Genus	Captures per trap		Percent total of captures	Captures per trap		Percent total of captures
			Control	Treated		Control	Treated	
Araneae			4.0	5.8	2.90	3.1	4.3	2.64
Acariformes			0.1	0.7	0.22	1.0	1.4	0.84
Isopoda			27.5	52.9	23.74	0.4	2.0	0.86
Cephalostigmata			0.0	0.1	0.01	0.0	0.1	0.02
Orthoptera			0.6	0.5	0.33	0.5	0.2	0.23
Embioptera			0.1	0.0	0.03	0.1	0.1	0.04
Blattodea			1.2	2.6	1.11	0.3	1.0	0.45
Mantodea			0.2	0.1	0.06	0.0	0.0	0.00
Isoptera			0.0	0.0	0.00	0.3	0.0	0.09
Hemiptera			1.9	2.8	1.39	3.3	6.5	3.50
Thysanoptera			0.1	0.4	0.13	0.0	0.0	0.00
Neuroptera			0.0	0.1	0.01	0.1	0.2	0.07
Lepidoptera			0.3	0.5	0.21	0.2	0.1	0.07
Diptera			2.9	6.7	2.82	1.9	1.7	1.26
Hymenoptera			80.8	141.1	65.58	114.8	133.4	89.05
	Formicidae		77.7	137.6	63.61	113.9	132.6	88.48
		Messor	16.0	30.8	13.82	16.3	37.4	19.26
		Plagiolepis	1.1	1.0	0.55	3.7	2.0	2.05
		Cataglyphis	41.1	65.9	31.61	21.6	41.3	22.56
		Pheidole	19.4	39.7	17.47	70.6	49.3	43.02
		Camponotus	0.3	0.0	0.07	0.0	0.0	0.00
		Tapinoma	0.0	0.0	0.00	1.5	1.3	0.97
		Crematogaster	0.1	0.2	0.09	0.3	1.5	0.63
Coleoptera			2.7	2.3	1.46	0.7	1.8	0.88

Classification according to Hickman et al., 1999.

Table 2Pathogenicity of *Beauveria bassiana* and *Metarhizium anisopliae* to *Tapinoma nigerrimum* workers.

	Mortality ^a , % (mean ± SE)	Kaplan–Meier survival análisis	
		AST ^b , days (mean ± SE)	95% CI
Control ^c	41.1 ± 8.8 a	16.9 ± 0.2 a	16.5–17.3
EABb 01/103-Su ^c	48.6 ± 8.5 a	15.1 ± 0.3 a	14.6–15.6
EAMa 01/58-Su ^c	64.7 ± 9.3 a	14.8 ± 0.3 a	14.3–15.3

^a Data in the same column followed by the same letter are not significantly different ($\alpha = 0.05$) according to the Tukey's HSD test.

^b Average survival time (AST) limited to 20 d. Data in the same column followed by the same letter are not significantly different ($\alpha = 0.05$) according to the log-rank test.

^c Control: 8 replicates with 100 ants per replicate. EABb 01/103-Su and EAMa 01/58-Su: 10 replicates with 100 ants per replicate.

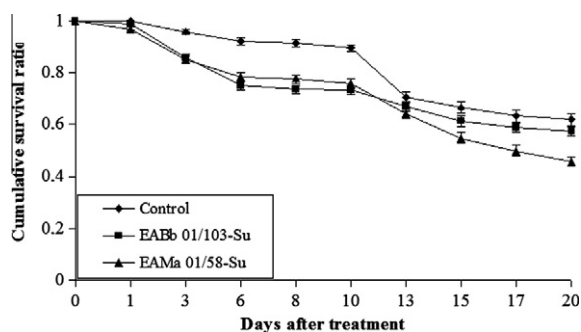


Fig. 3. Cumulative survival ratio (mean ± SE) of *Tapinoma nigerrimum* adults exposed to soil treated with *Beauveria bassiana* (EABb 01/103-Su) and *Metarhizium anisopliae* (EAMa 01/58-Su) or untreated soil.

The mean values ranged between 41.1% and 64.7%. The average survival times (AST) of each treatment are shown in Table 2. No significant reduction in AST was observed between treatments (from 16.9 days in control to 15.6 days in treatment with *B. bassiana* and 14.8 days in treatment with *M. anisopliae*). The fungal treatment did not cause a significant reduction of the survival ratio of ants when compared to the control (Fig. 3). The three curves progress very similarly showing a decrease between 10 and 13 days.

3.3.2. Activity of *T. nigerrimum*

In both controls and fungally challenged ants, significant differences were detected in their activity before and after treatment ($F_{1,25} = 18.07$, $P < 0.001$), with an activity decrease over time. The workers through the linker tube counted before the treatment for control, *B. bassiana* isolate and *M. anisopliae* isolate were 4.3, 3.9 and 4.8, while after treatment were 3.6, 4.1 and 3.5 respectively. Nonetheless, the fungal treatment had no detectable effect on ants ($F_{2,25} = 0.256$, $P = 0.78$).

3.3.3. Fungal dispersal by *T. nigerrimum*

Soil samples from the annex artificial nests were collected and examined by the CFU method to determine the presence of fungi. In all plates, the absence of EF was observed. This finding was caused by a lack of inoculum dispersal by the ants.

4. Discussion

The pitfall collections represented a broad diversity of olive orchard dwelling arthropods. The olive grove supported an abundant group of edaphic arthropods that was numerically dominated by ants. In accordance with previously reported results, the composition of soil fauna in the Iberian Peninsula is similar to that obtained in our study with the most abundant taxa being Formici-

dae (Morris and Campos, 1999; Santos et al., 2007b). All ants genera collected are frequently found in olive orchards nesting under the tree canopy or exclusively in open zones (Redolfi et al., 1999). The results of the two applications to the same field plot showed that the EAMa 01/58-Su isolate, which have been recovered up to 10^5 CFU g^{-1} soil during the experiment, did not have a significant effect on the arthropods as there were more individuals collected in the treatment subfield than in the control subfield during both years. Furthermore, there were no infected individuals. Parker et al. (1997) performed a *B. bassiana* soil application and obtained 5% infected individuals. However, it was not possible to confirm whether the infected arthropods were a result of the treatment or due to the natural presence of *B. bassiana* in that forest soil.

The pathogenicity of EF against ants was performed with the *T. nigerrimum* species. Although it is considered one of the most abundant olive ants (Morris and Campos, 1999; Redolfi et al., 1999), it was not found into pitfall traps, due to seasonality of this species (Retana et al., 1990); in Southern Spain is mainly found in spring time while the experiment was performed in summer. Several species of ants are susceptible to EF (Siebeneicher et al., 1992; Rodrigues et al., 2010), but no effects were detected for any of the fungi evaluated under the conditions of this study. Ants have evolved many effective defense mechanisms against pathogens. One of these common defenses used by ants is grooming. Ants spend more time on grooming (self and allo-grooming) in the presence of fungi as they remove the conidia directly from the exoskeleton (Siebeneicher et al., 1992; Richard and Errard, 2009; Reber et al., 2011). Furthermore, entomopathogenic fungi can be inhibited by antimicrobial secretions of the ants' metapleural and mandibular glands (Bot et al., 2002; Rodrigues et al., 2009). Beattie et al. (1986) suggested that these secretions are not decisive in the survival of the host while the innate immune system or inducible antibiotic peptides in the haemolymph are key factors impairing the fungal infection. Ants are known to be able to carry conidia of EF on the cuticle or infrabuccal pellets (Bird et al., 2004; Pagnocca et al., 2008), which is an important factor in the dispersal of this fungi. A lack of significant interaction between ants and EF could be attributed to the previously mentioned mechanisms. However, it was previously reported as an antagonist of non-sterile soil (Pereira et al., 1993). Soil antagonism against EF was further evidenced by the lack of development of the fungus on fire ants killed (Pereira et al., 1993) or in soil (Siebeneicher et al., 1992). This antagonism may affect conidia in the soil by inhibiting germination and penetration to insects.

Based on the lack of infection among the trapped arthropods and the low level of infection among treated nests, our results suggest that these EF could be applied to olive orchard soil without a significant negative direct or indirect impact on the soil dwelling arthropod population.

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